Highlights

Retrieval-augmented large language models for training-free trajectory reconstruction under satellite navigation spoofing in aviation and maritime surveillance

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- Training-free trajectory reconstruction after satellite-navigation spoofing
- One engine serves aircraft and ships; only prompt, index and trusted signal change
- Ship trajectories reconstructed with a frozen language model and no ship training
- Statistically indistinguishable from a trained, ship-specific network in every phase tested
- Significantly beats every classical, recurrent and retrieval-only baseline; validated on 1,544 flights and 60 vessels against fourteen maritime baselines

Retrieval-augmented large language models for training-free trajectory reconstruction under satellite navigation spoofing in aviation and maritime surveillance

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Abstract

When an aircraft or a vessel is suspected of receiving spoofed satellite-navigation signals, surveillance operators must discard its broadcast positions yet keep tracking it. Existing trajectory predictors are trained for a single transport domain and must be rebuilt for each new one. This paper extends TrajRAG (Trajectory Retrieval-Augmented Generation) from aviation to a domain-agnostic framework. *Contribution in artificial intelligence:* we formulate post-detection trajectory reconstruction as retrieval-conditioned sequential inference by a large language model whose parameters are never updated, and we isolate a domain adapter – a textual-state prompt template, a phase-indexed vector memory, a trusted non-satellite signal and a data-calibrated kinematic feasibility bound – as the only component that changes between domains. *Application in engineering:* one reconstruction engine maintains track continuity for air-traffic surveillance based on Automatic Dependent Surveillance–Broadcast messages and for vessel traffic services based on Automatic Identification System messages. Transferred to shipping without any model training, the framework is evaluated on commercial vessels in two United States port corridors, against a retrieval memory that structurally excludes both the test set and any chronologically later

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voyage, and against fourteen maritime baselines. It reduces mean reconstruction error by a factor of 1.3–1.5 relative to the best classical baseline in every voyage phase, significantly so (paired Wilcoxon test, $p < 10^{-3}$ in every case). Against a trained, ship-specific convolutional network, it is statistically indistinguishable in every phase (Houston, $n = 50$), matching – without any maritime training data – a model purpose-trained on this corridor's own traffic. A complete blackout reconstruction finishes in under 35 s, well within the 5-minute vessel reporting interval. These results show that prompt and memory design, rather than retraining, can carry trajectory reconstruction across transport domains.

Keywords: Satellite navigation spoofing, Trajectory reconstruction, Retrieval-augmented generation, Large language models, Automatic Identification System, Air traffic surveillance, Cross-domain transfer

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1. Introduction

Aircraft and ships are tracked by cooperative surveillance systems that rebroadcast positions computed by an on-board Global Navigation Satellite System (GNSS) receiver. In aviation the carrier is Automatic Dependent Surveillance–Broadcast (ADS-B) (Federal Aviation Administration, 2007; Khan et al., 2025); at sea it is the Automatic Identification System (AIS) (International Telecommunication Union, 2014). Neither channel authenticates the position it carries, so a receiver that is jammed or spoofed passes the corrupted position directly to air traffic control (ATC) and vessel traffic services (VTS). Interference has become a routine operational condition rather than a rare event. Aviation monitoring reports a sustained rise in GNSS anomalies in and around conflict zones (Aireon LLC, 2025; Felux et al., 2025; Ieropoulos, 2024; Lo et al., 2025). Maritime trackers reported roughly 700 jamming and spoofing events per day across the global fleet in 2024 (Windward, 2025). In the Baltic Sea, the number of affected vessels grew from about 1,225 in the first quarter of 2025 to more than 5,800 in the second (GPSPATRON, 2025). A single two-week window in mid-2025 recorded more than 12,000 spoofing incidents involving over 3,000 vessels (Kpler, 2025), and sustained spoofing has been documented in the Persian Gulf (Skuld, 2025; The Maritime Executive, 2025).

Detecting interference is necessary but not sufficient. Once a detector flags an aircraft or a vessel, its satellite-derived positions must be discarded,

yet the operator still has to deconflict traffic, enforce separation and verify that the asset stays clear of restricted areas. In a narrow ship channel or a busy terminal area, losing a reliable track for even half an hour is operationally significant. The engineering problem is therefore *post-detection reconstruction*: estimating where the asset is from its last trusted state, from signals that do not depend on satellites, and from what similar assets have done before.

Today this gap is usually filled by trajectory predictors used as extrapolators. These predictors are almost always domain-specific. Aircraft predictors are trained on flights (Pang et al., 2019; Kwak and Lee, 2025), and vessel predictors are trained on AIS voyages (Li et al., 2023; Shin et al., 2024; Nguyen and Fablet, 2024). Each must be rebuilt, retuned and revalidated when it is moved to a new region, a new traffic mix or a new transport mode. For an organisation that operates both airspace and coastal surveillance, or for a vendor that supplies both, this means maintaining two unrelated model families for what is essentially the same task.

In the conference version of this work (Felendler et al., 2026) we introduced TrajRAG for aircraft. TrajRAG converts trajectory segments into textual state descriptions and stores them in phase-specific vector indices. During a blackout it retrieves phase-consistent historical segments and lets a large language model (LLM) propose the next position, which is accepted only inside a kinematic feasibility envelope. That study was limited to aviation: it did not establish whether the design is specific to aircraft dynamics and ADS-B, or whether it is a reusable reconstruction method. This paper answers that question. Our thesis is that *nothing in the reconstruction engine needs to be learned per domain*: moving to a new transport mode should require only a new prompt template, a new retrieval index and a small set of physically motivated constants, all derived from domain data by fixed procedures.

We test this thesis on shipping, which differs from aviation in almost every respect that matters to a predictor. Vessels are an order of magnitude slower. AIS is sampled irregularly. Phase dynamics are dominated by berthing manoeuvres rather than altitude changes, the environmental drivers are currents and waves rather than winds aloft, and the GNSS-independent signal available on board is different. If the same engine, with its LLM frozen and no maritime gradient update, competes with dedicated maritime models, the result is evidence that retrieval and prompting can replace retraining in this class of engineering problem.

Contributions. The contributions of this paper are:

1. **A domain-agnostic formulation (AI contribution).** We separate post-detection reconstruction into a shared engine (retrieval, frozen-LLM proposal, feasibility guard and feedback loop) and a domain adapter of eight elements. We specify the adapter formally and show that it is the only part that changes between aviation and shipping (Section 3).
2. **A maritime instantiation built without training.** We construct the maritime adapter (AIS state schema, gyrocompass heading as the trusted signal, departure/transit/arrival phases, marine environmental context and a voyage prompt template), together with a new public-data corpus of 3,506 voyages from 26.2 million AIS reports in two United States port corridors (Sections 4 and 5.1).
3. **A rigorous maritime benchmark.** We compare against fourteen baselines: classical, recurrent, retrieval-only, seven modern forecasters and AIS-ACNet, a trained ship-specific network published in this journal (Shin et al., 2024). All are run on a common held-out test set with paired significance tests and distance- and time-normalised errors (Section 6).
4. **A transferable calibration procedure for the feasibility guard.** We show that the procedure used to calibrate the guard transfers across domains, while the constant it produces does not. Vessel dynamics give a phase-dependent displacement-ratio distribution with no counterpart in aviation (Section 3.5).
5. **A real-time dual-domain system design.** We describe how one deployment serves ATC and VTS, and we show that the reconstruction latency fits inside the reporting interval of both surveillance channels (Sections 3.6 and 6.12).

Relation to the conference version. The conference paper (Felendler et al., 2026) covers aviation only. The domain-adapter formulation, the maritime instantiation and corpus, the full maritime evaluation (fourteen baselines, normalised errors, significance tests and corridor analysis), the cross-domain calibration analysis and the dual-domain system architecture are new. Aviation results are summarised in Section 6.1 only as the reference point for the cross-domain comparison; we do not repeat the aviation ablations, backbone study or real-world case studies, and refer the reader to the conference paper for them.

The rest of the paper is organised as follows. Section 2 reviews related work. Section 3 presents the domain-agnostic framework and the system architecture. Section 4 describes the two instantiations. Section 5 details the experimental design, and Section 6 reports the results. Section 7 discusses implications and threats to validity, and Section 8 concludes.

2. Related work

2.1. Satellite-navigation interference in transport surveillance

ADS-B horizontal position and velocity are derived from the aircraft's GNSS receiver, so interference propagates directly into the surveillance picture (Khan et al., 2025; Pik et al., 2024, 2025). AIS has the same weakness: position, SOG and COG are GNSS-derived and broadcast over an unauthenticated VHF link (International Telecommunication Union, 2014). Most research addresses *detection*. In aviation, studies correlate ADS-B position gaps and integrity indicators with notices to airmen (Pik et al., 2024). At sea, recent work uses AIS as a sensor of opportunity to detect and localise interference over wide areas (Park et al., 2026; Winkel et al., 2026), and recurrent classifiers flag spoofed AIS points (Raj and Kumar, 2025). These detectors decide *when* a track can no longer be trusted. Our work starts at that point and addresses what the operator sees *after* the decision.

2.2. Trajectory prediction and reconstruction

Classical trackers such as Kalman and interacting-multiple-model filters (Kalman, 1960; Ren et al., 2024) work well over short horizons. Under interference, however, corrupted measurements are discarded rather than smoothed, so the filter runs open loop and its error grows with the gap. Autoregressive models behave similarly. Recurrent networks (Pang et al., 2019; Hashemi et al., 2024) capture non-linear motion but depend on training coverage and drift without feedback. Recent reviews of aircraft (Kwak and Lee, 2025) and ship trajectory prediction (Li et al., 2023; Guo and Ma, 2026) describe a mature but domain-siloed literature dominated by recurrent, convolutional, attention-based and hybrid networks. Among maritime models, AIS-ACNet (Shin et al., 2024) couples parallel dilated-convolution encoders for position and for auxiliary SOG/COG targets and was evaluated on Port of Houston AIS. Because it predicts forward only and has public code, it is the most suitable literature baseline for our protocol. TrAISformer (Nguyen and Fablet, 2024) discretises AIS states and predicts them

with a transformer. Generic forecasters such as DLinear (Zeng et al., 2023), PatchTST (Nie et al., 2023), iTransformer (Liu et al., 2024), Informer (Zhou et al., 2021), Autoformer (Wu et al., 2021), FEDformer (Zhou et al., 2022) and TimesNet (Wu et al., 2023a) provide strong domain-independent baselines, but each must be trained on target data.

A separate line of work imputes ordinary AIS coverage gaps. MH-GIN (Liu et al., 2026a) uses multi-scale heterogeneous graphs. HABIT (Spiliopoulos et al., 2026) uses historical hexagon transitions and path search. VISTA (Liu et al., 2026b) lets an LLM choose an imputation pattern from a knowledge graph built from historical counts, which gives each repair a readable provenance. These methods are the closest to ours in spirit, but they interpolate between two known endpoints and address benign gaps. Our setting is forward-only: the end of the blackout is unknown, and the positions inside it are deliberately discarded.

2.3. Large language models and retrieval for temporal data

LLMs can forecast numeric series zero-shot when values are serialised as text (Gruver et al., 2023), and they can be reprogrammed for forecasting with a small trained adapter (Jin et al., 2024). Retrieval-augmented generation grounds a parametric model in non-parametric memory (Lewis et al., 2020), and TS-RAG (Ning et al., 2025) shows that retrieval improves zero-shot time-series foundation models. For flights, LLMs have been fine-tuned on serialised ADS-B states (Luo and Zhou, 2025) and assessed qualitatively for reconstruction (Zhang and Mott, 2024). None of these approaches encodes the constraints that make trajectory reconstruction safe to deploy: operational phase, a trusted signal that is physically independent of the corrupted one, and a hard bound on feasible displacement. None of them has been shown to move between transport domains without retraining.

2.4. Positioning

Table 1 summarises where this work sits. Its distinguishing combination is a post-detection, forward-only formulation, no target-domain training, an explicit physical guard and a demonstrated transfer across two physically distinct transport domains.

Table 1: Positioning of this work relative to representative trajectory-prediction and reconstruction approaches. "Post-detection" means that horizontal positions after the onset are withheld rather than denoised. "No training" means that no model parameters are fitted on target-domain trajectories.

Approach	Post-det.	No training	Phys. guard	Multi-domain	Interpretable
Kalman / IMM filters	✓	✓	–	✓	–
Recurrent/conv. predictors	–	–	–	–	–
Generic time-series forecasters	–	–	–	–	–
AIS imputation (MH-GIN, VISTA, HABIT)	–	partly	–	–	partly
Fine-tuned LLM flight prediction	–	–	–	–	✓
Zero-shot/retrieval LLM forecasting	–	✓	–	✓	–
TrajRAG, aviation only (Felendler et al., 2026)	✓	✓	✓	–	✓
This work	✓	✓	✓	✓	✓

3. Domain-agnostic reconstruction framework

3.1. Problem formulation

Let $X = \{x_0, \ldots, x_n\}$ be a time-ordered trajectory of an aircraft or a vessel. Each state $x_i = (y_i, k_i, a_i, \text{time}_i)$ contains a horizontal position $y_i = (\phi_i, \lambda_i)$, kinematic fields k_i (for example ground speed and track, or SOG and COG) and the reading a_i of a trusted auxiliary signal. The trusted signal must be *physically* independent of the satellite receiver, not merely reported in a separate field. An external detector raises an alarm at index t. From then on, every satellite-derived field (y_i and the satellite-derived part of k_i, for $i > t$) is withheld from the reconstruction, while a_i remains available. The task is to produce

$$\hat{Y}_{t+1:r} = \{\hat{y}_i\}_{i=t+1}^{r} \tag{1}$$

until trusted reporting resumes, or the trip ends, at index r. The available information is the verified prefix $X_{0:t}$, the trusted signal $a_{0:r}$, environmental context obtained independently of the asset, and historical trajectories of

the same domain. This is not denoising, because corrupted observations are dropped rather than filtered. It is not interpolation either, because the end state is unknown. We evaluate it as forward-only reconstruction against the true, uncorrupted positions.

3.2. Shared engine and domain adapter

We factor the method into a shared engine and a domain adapter. The *engine* comprises a frozen text-embedding model g, a frozen LLM f, an inner-product vector index, the iterative propose–validate–update loop of Algorithm 1, the feasibility guard and a fixed output schema (one latitude/longitude pair per step). No component of the engine is trained, fine-tuned or adjusted for a domain. A domain adapter is the tuple

$$\mathcal{D} = \langle \mathcal{S}, a, \mathcal{P}, \pi, \tau, \mathcal{E}, \mathcal{M}, \kappa \rangle, \tag{2}$$

whose elements are:

- $\mathcal{S}$, the *state schema*, which lists the fields of a surveillance report and marks each as satellite-derived (untrusted after onset) or trusted;
- a, the *trusted auxiliary signal*;
- $\mathcal{P}$ and π, the *phase taxonomy* and a phase-assignment rule that uses only trusted information;
- τ, the *verbaliser*, a prompt template that turns a trajectory segment, the trusted signal and the environmental context into a short textual state description;
- $\mathcal{E}$, the *environmental context provider*, queried at the last accepted (trusted or reconstructed) position and time;
- $\mathcal{M} = \{\mathcal{M}_p\}_{p \in \mathcal{P}}$, the *phase-indexed memory*, built by applying τ and g to historical trajectories of the domain;
- $\kappa = (\Delta t, \{\alpha_p\}, \text{SPEEDRULE})$, the *kinematic constants*: the sampling interval, per-phase tolerance factors and the rule that updates the reference speed during a blackout.

Algorithm 1 Online reconstruction with the shared engine and a domain adapter $\mathcal{D}$

Require: verified prefix $X_{0:t}$; trusted signal $a_{0:r}$; adapter $\mathcal{D} = \langle \mathcal{S}, a, \mathcal{P}, \pi, \tau, \mathcal{E}, \mathcal{M}, \kappa \rangle$ with $\kappa = (\Delta t, \{\alpha_p\}, \text{SPEEDRULE})$; frozen LLM f; frozen encoder g; retrieval size K

Ensure: reconstructed positions $\hat{Y}_{t+1:r}$

$p \leftarrow \pi(X_{0:t}, a_{0:t})$ ▷ phase from trusted information only

$(\hat{y}_t, v) \leftarrow (y_t, v_t)$ ▷ last verified position and speed

$q \leftarrow \tau(X_{0:t}, a_{0:t}, (\hat{y}_t, t))$ ▷ textual state of the prefix

for $i = t+1, \ldots, r$ **do**

  $C \leftarrow \text{TopK}(g(q), \mathcal{M}_p, K)$ ▷ cosine similarity

  $\tilde{y}_i \leftarrow f(q, C, a_i, (\hat{y}_{i-1}, i))$ ▷ fixed schema: latitude, longitude

  $d_{\max} \leftarrow \alpha_p \, v \, \Delta t$

  if $d_{\text{hav}}(\tilde{y}_i, \hat{y}_{i-1}) > d_{\max}$ **then**

    $\tilde{y}_i \leftarrow \text{Project}(\tilde{y}_i, \hat{y}_{i-1}, d_{\max})$; raise low-confidence flag

  end if

  $\hat{y}_i \leftarrow \tilde{y}_i$

  $v \leftarrow \text{SPEEDRULE}(v, \hat{y}_{i-1}, \hat{y}_i, i)$ ▷ domain-specific (Section 4)

  $q \leftarrow \tau(X_{0:t} \cup \hat{Y}_{t+1:i}, a_{0:i}, (\hat{y}_i, i))$ ▷ feedback: re-query from reconstructed prefix

end for

return $\hat{Y}_{t+1:r} = \{\hat{y}_i\}_{i=t+1}^{r}$

Three of these elements are semantic choices that a domain expert makes once: the trusted signal, the phase taxonomy and the vocabulary of the verbaliser. The remaining elements are data artefacts produced from the domain corpus by procedures that are identical in both domains (Sections 3.3 and 3.5). Porting the method to a new domain therefore requires a prompt template, an index and a calibration run. It requires no training run.

3.3. Offline memory construction

Historical trajectories are cleaned (duplicates removed, timestamps made monotonic), resampled to the adapter's Δt when the native sampling is irregular, and segmented into phases. Each segment is verbalised by τ into a description of its spatial evolution, kinematic trend, phase, elapsed time in phase and aligned environmental context. Timestamps are expressed as time from the start of the phase so that segments recorded on different days are

comparable. Descriptions are embedded by g, L2-normalised and stored in a separate inner-product index per phase (and, in the maritime adapter, per corridor). Each index entry keeps a pointer to the numeric continuation of its segment, which is shown to the LLM at inference time. Separating indices by phase prevents incompatible motion regimes, such as berthing and open-channel transit, from being retrieved together. New data are added by appending entries, without re-fitting any model.

3.4. Online retrieval-conditioned reconstruction

At onset, the phase is determined by π from trusted information, and the verified prefix is verbalised into a query. At each step the engine retrieves the K most similar segments from $\mathcal{M}_p$ by cosine similarity. The LLM then receives the query, the retrieved segments with their continuations, the trusted reading a_i and the environmental context, and returns the next position in the fixed schema. After the position passes the guard, the query is rebuilt from the extended (verified + reconstructed) prefix, so the retrieved neighbours evolve as the reconstruction progresses (Algorithm 1). The LLM thus acts as a retrieval-conditioned *proposal generator*: it adapts and blends the retrieved precedents to the current context, but it never has the final word on feasibility.

3.5. Kinematic feasibility guard and its calibration procedure

Each proposal $\tilde{y}_i$ must lie within a disc centred at the last accepted position, with radius

$$d_{\max} = \alpha_p \, v \, \Delta t, \tag{3}$$

where v is the current reference speed and α_p is a phase tolerance. Proposals outside the disc are projected onto its boundary along the bearing from the previous position, and the step is flagged as low confidence. The guard therefore turns any LLM error, including a hallucinated position, into a bounded displacement.

The tolerance is not tuned on reconstruction accuracy. It is set by a fixed, accuracy-independent *calibration procedure*. On the domain's training trajectories we compute the step displacement ratio

$$\rho_i = \frac{d_{\mathrm{hav}}(y_{i-1}, y_i)}{v_{i-1} \, \Delta t}, \tag{4}$$

excluding steps with near-zero reference speed, and we set α to a high quantile of ρ (the 99th percentile in this study). For aircraft this procedure gives

$Q_{0.99}(\rho) = 1.18$ and hence $\alpha = 1.2$ (Felendler et al., 2026). Section 6.9 applies the same procedure to vessels and shows why the procedure, rather than the constant, is what transfers between domains.

3.6. Real-time dual-domain system architecture

Fig. 1 shows how the framework is deployed as one service for both surveillance domains. ADS-B and AIS streams feed an external spoofing detector, which this paper treats as a given interface. When an alarm is raised for an asset, a trust gate stops forwarding that asset's satellite-derived fields. The message type selects the domain adapter. The shared engine then produces one reconstructed position per reporting interval until the detector clears the alarm or the trip ends. Each reconstructed position carries a confidence flag that records whether the guard had to clamp the proposal, so a controller or VTS operator sees the reconstruction together with an explicit indication of how far it relies on the physical bound rather than on precedent. Because adapters contain only prompts, indices and constants, a new domain or corridor is added by deploying a new adapter. The engine, and any accreditation evidence gathered for it, stays unchanged. Memories are refreshed by appending completed trips, for example once a day. Section 6.12 checks the latency of this loop against the reporting interval of each channel.

4. Domain instantiations

Table 2 lists both adapters side by side.

4.1. Aviation adapter

The aviation adapter is that of the conference version (Felendler et al., 2026) and is summarised here only for completeness. The trusted signal is barometric altitude, which comes from the air-data system rather than the GNSS receiver. Phases (takeoff, cruise, landing) are assigned from altitude trends. The verbaliser describes the flight phase, time since takeoff, altitude trend, heading and speed trend and the aligned wind and turbulence indicators. During a blackout, ground speed and heading are not read from ADS-B; they are inferred from consecutive accepted reconstructed positions and used as the reference speed of Eq. (3).

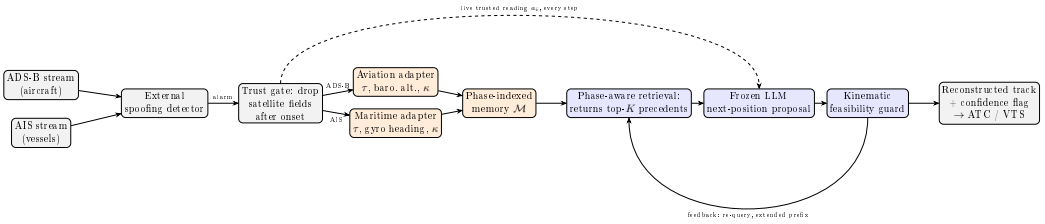


Figure 1: Architecture of the real-time, domain-agnostic system. Both surveillance streams pass through the same detector interface and trust gate. A domain adapter (orange) supplies the domain-specific parts: the textual-state prompt template, the trusted non-satellite signal, the kinematic constants, and the phase-indexed memory $\mathcal{M}$ built from that domain's history – shown as a separate node from the shared engine's retrieval step (blue), which searches $\mathcal{M}$ and returns only the top-K precedents for the current query, not the memory itself. The dashed arrow shows the trusted reading a_i (heading or barometric altitude) entering the loop live at every step, independently of the one-shot adapter routing above it. The engine asks a frozen large language model for the next position and accepts it only inside the kinematic envelope; the accepted position extends the verified prefix, and the query is rebuilt from it before the next retrieval. ATC: air traffic control; VTS: vessel traffic service.

4.2. Maritime adapter

Trusted signal. On commercial vessels, true heading is normally supplied by a gyrocompass, which is independent of GNSS, whereas SOG and COG are computed by the GNSS receiver. We therefore treat heading as the maritime counterpart of barometric altitude, and SOG and COG as untrusted after onset. Heading was present on 98–99% of AIS reports in our corpus, comparable to the availability of barometric altitude in ADS-B. Vessels whose heading comes from a satellite compass break this independence assumption. We return to this point in Section 7.4.

Phases and memory. Voyages are segmented into departure (leaving the berth and manoeuvring), transit (steady channel or approach passage) and arrival (slowing and berthing). One index is built per phase and per corridor, so the adapter's memory for a port is simply a small set of indices built from that port's history. This segmentation is computed offline from each completed voyage's full SOG trend, which is the practical way to build phase-indexed memory from historical data; it is not an implementation of the online phase-assignment rule π of Section 3, which is restricted to trusted information available up to the current step. At evaluation time, each benchmark voyage's phase label is taken from this offline segmentation, so the

maritime instantiation evaluates reconstruction *within* a known phase rather than online phase identification; we return to this in Section 7.4.

Verbaliser. The voyage template mirrors the flight template but uses maritime vocabulary and drivers. A representative departure state reads: "Vessel type: cargo. 15.0 minutes into the departure phase, the vessel is tracking northeast through the Galveston Bay entrance channel, heading 048° with course over ground 045° and speed over ground 8.4 knots, gradually accelerating away from berth. Sea state is moderate, with 0.6 m wave height and a light following swell; a moderate current of 0.9 km/h runs with the vessel's heading, and a 12 km/h wind blows from the southeast. Motion has been steady, with no abrupt heading change." The environmental provider attaches wave height, direction and period, swell height, ocean-current speed and direction, sea-surface temperature and 10 m wind speed and direction from Open-Meteo Marine (Open-Meteo, 2025). Appendix A gives both templates verbatim, together with a fully worked maritime reconstruction step.

Sampling. Unlike ADS-B, AIS reporting is irregular: Class A transponders change their rate with speed and navigational status, and messages are lost to channel contention (International Telecommunication Union, 2014). Rather than letting the guard handle a variable Δt, the adapter resamples every voyage onto a uniform 5-minute grid before segmentation. We verified that all consecutive points within a segmented phase are exactly 300 s apart. This keeps the engine identical across domains, since the irregularity is absorbed by the adapter's preprocessing.

Reference-speed rule. The maritime SPEEDRULE holds the reference speed v of Eq. (3) at the last *trusted* SOG – the value recorded immediately before onset – for the entire blackout, rather than updating it from any post-onset field. This is the deployable rule used throughout Section 6: it uses no information that a real blackout would withhold. An earlier implementation instead updated the reference speed from the recorded SOG at each reconstructed step; because SOG is GNSS-derived, that variant used information a genuine blackout would not provide, and we do not report it as a result in this paper. Section 6.13 reports the corrected, deployable numbers together with the size of the effect relative to that earlier, non-deployable variant, as a transparency check rather than as an additional claimed result.

5. Experimental design

5.1. Data

Table 3 summarises both corpora. The aviation corpus is described in Felendler et al. (2026). The maritime corpus is new. We downloaded 44 days of raw AIS position reports (19 April–1 June 2025, the same calendar window as the aviation collection) from the NOAA Marine Cadastre (NOAA Office for Coastal Management, 2025), a public United States government archive with no access restrictions, which supports replication. Two corridors with contrasting geometry were selected. The Houston Ship Channel, from the Galveston Bay entrance to the Port of Houston, is a long, narrow inland channel with dense tanker and cargo traffic. The Los Angeles/Long Beach approach is a shorter, more open approach. From 26.2 million reports we reconstructed 3,506 cargo and tanker voyages, segmented them into phases and attached the marine environmental context. For testing we randomly selected 250 vessels (125 per corridor) with at least four 5-minute points in every phase, the same minimum segment length as in aviation; selection is a uniform random sample over the full collection window, with no chronological holdout by itself. AIS-ACNet's training and validation sets exclude all test voyages and are themselves chronologically split (Section 5.3).

An earlier implementation of TrajRAG's retrieval had two problems. First, it excluded only the exact query voyage from a corridor's corpus-wide index, not the other 124 test voyages, so a query could retrieve another test voyage as a precedent – retrieval memory contamination, not a disclosable design choice. Second, the memory had no chronological cutoff, so a query could retrieve a precedent recorded after it, which a real-time deployment could never do. We rebuilt the memory to fix both: a corridor's retrieval index is now built only from the chronologically earliest half of its non-test voyages (1,141 of 2,283 non-test Houston voyages; 486 of 973 non-test Los Angeles/Long Beach voyages), and retrieval additionally excludes every voyage in the test set, not only the query. Every TrajRAG reconstruction reported in this paper queries only this pre-cutoff memory, so no retrieved precedent can postdate the query, by construction rather than by an exclusion list.

Reconstructing under the corrected memory, and under the corrected preprocessing of Section 5.1, requires fresh LLM calls for every query voyage, which this paper's budget could not extend to the full 125-per-corridor test set together with the ablation and matched-forecaster experiments of Sections 6.10 and 5 below. We therefore report TrajRAG's headline maritime

results on a smaller, still randomly-drawn common test set: 50 Houston and 10 Los Angeles/Long Beach voyages, sampled from the subset of the original 125-per-corridor test voyages that postdate each corridor's memory cutoff (68 and 67 such voyages were available, respectively). Every other method in Table 8 and Table 16 – Kalman, ARIMA, LSTM, BiLSTM, Mean-Retrieval, Numeric-kNN and AIS-ACNet – is rescored on this identical 50/10-voyage set for a matched comparison; AIS-ACNet is additionally reported on the full 125-voyage set as a robustness check that the smaller matched sample is not anomalous (Section 6). Section 7.4 discusses the resulting loss of statistical power, particularly at Los Angeles/Long Beach.

Preprocessing correction. The 5-minute resampling grid originally interpolated latitude and longitude with a time-weighted method that could use an AIS report recorded *after* a given grid timestamp, including the single most load-bearing point in the whole pipeline: the last point of the verified prefix that every method treats as ground truth to reconstruct from. Every other resampled field (SOG, COG, heading, ...) already used a causal last-observation-carried-forward rule; we moved latitude and longitude onto the same rule, removing the leak. Rescoring the deterministic classical and numeric baselines (Kalman, ARIMA, Numeric-kNN) on the corrected data versus the original shows this is not a negligible effect: every one of 12 method/phase/corridor cells we checked became worse under the correction, by 2–40%, confirming the leak had been making reconstruction look easier than it is. All results in this paper use the corrected data.

5.2. Blackout protocol

We simulate post-detection blackouts on real, uncorrupted trajectories, so that the true positions are available as ground truth. Following a safety-oriented, attack-agnostic stance, we model every attack (gradual drift, velocity manipulation or capture-and-pull-off) the same way: all satellite-derived fields after onset are withheld. In aviation, onset times are randomised and blackouts last 5–15 steps (10–30 min) (Felendler et al., 2026). In the maritime benchmark, each phase segment of N points is split at $\lfloor N/2 \rfloor$. The first half is the verified prefix, and the second half is withheld and must be reconstructed. This gives mean blackout durations of 37.2, 35.2 and 66.1 min for departure, transit and arrival on the Houston test set. The maritime protocol is applied identically to every maritime method, including every baseline: no baseline receives any field reported after the onset index. Because the two

domains use different protocols and sampling intervals, absolute errors are compared within a domain, not across domains.

5.3. Baselines

Classical, recurrent and retrieval-only. These are a Kalman filter (a constant-velocity model that accounts for currents in the maritime case, using only prefix data and environmental context), ARIMA fitted to each verified prefix, LSTM and BiLSTM networks trained on the non-test voyages of each phase (about 1,600–3,300 per phase), and Mean-Retrieval, which averages the continuations of the top-K retrieved segments without an LLM.

Numeric-kNN. This baseline is a non-parametric retrieval model operating on numbers rather than text. It finds the five training prefixes of the same phase closest to the test prefix by mean point-wise Haversine distance and predicts each withheld step as the coordinate-wise mean of their continuations. It isolates the value of textual states combined with LLM blending over plain numeric nearest neighbours.

AIS-ACNet. We retrain AIS-ACNet (Shin et al., 2024) from its public implementation on our voyages (on the corrected data of Section 5.1), using a chronological 85/15 train/validation split of the non-test Houston voyages, and evaluate it on the common test set with the same blackout boundary. We report it both on the full 125-voyage test set and on the 50/10-voyage set TrajRAG is evaluated on, to check that the smaller matched sample is not anomalous (Section 6). AIS-ACNet's convolutional receptive field is fixed, so we train one model per phase, with the window set near the 95th percentile of the true half-phase length (Table 4). Voyages whose true withheld half exceeds the window are truncated to it for both the model's output and the scoring mask – 4, 6 and 5 of the 125 test voyages in departure, transit and arrival respectively (3–5% per phase); shorter ones are padded with edge values and masked out of the loss and the metric. For the remaining 95–97% of voyages per phase, AIS-ACNet reconstructs exactly the same horizon as TrajRAG; for the truncated tail, AIS-ACNet is scored only on the first *window* steps of the blackout, not the full one TrajRAG reconstructs.

Modern forecasters. DLinear, PatchTST, iTransformer, Informer, Autoformer, FEDformer and TimesNet are trained with the Time-Series-Library implementations (Wu et al., 2023b) on eight features per step (latitude, longitude, SOG, COG, heading, wave height, 10 m wind speed and current speed). The features are z-scored with training-split statistics. Encoder–decoder models receive the tail of the *observed* window as their decoder warm-up,

so no target value leaks into the input. These architectures need fixed input and output lengths. Following the aviation harness of the conference paper, we use a small exact-fit window per phase (4, 3 and 4 steps for departure, transit and arrival) and include a voyage only if its half-phase length matches the window. This retains about 80%, 71% and 77% of eligible voyages. The forecasters therefore solve an easier, shorter-horizon problem on a subset of voyages, and Table 8 reports them in a separate block.

5.4. Metrics and statistical testing

The primary metric is the mean Haversine error (MHE) (Zhao et al., 2024) over the withheld steps T,

$$\mathrm{MHE} = \frac{1}{|T|} \sum_{i \in T} d_{\mathrm{hav}}(\hat{y}_i, y_i), \tag{5}$$

reported per phase. A given absolute error means more for a slow ship than for a fast one, so we also normalise MHE by two quantities that are independent of the method and computed from ground truth: the mean distance travelled during the blackout and the mean blackout duration. Differences between TrajRAG and each full-horizon baseline are tested with paired two-sided Wilcoxon signed-rank tests (Wilcoxon, 1945) on per-voyage errors of the same voyages (50 Houston, 10 Los Angeles/Long Beach). We report raw p-values and note which conclusions survive a Bonferroni correction. We additionally report, for each comparison, the median paired difference (baseline minus TrajRAG) with a 95% bootstrap confidence interval over 10,000 resamples of voyages, and the median, 90th percentile and maximum of the per-voyage error distribution for every full-horizon method.

5.5. Implementation

Both instantiations use `text-embedding-3-large` as g and FAISS inner-product indices (Johnson et al., 2021). The aviation adapter uses `gpt-5.1` with $K = 5$, the configuration selected in the conference study. The maritime adapter uses `gpt-4.1` with $K = 6$, fixed before any maritime test-set result was observed; neither backbone was fine-tuned, and no maritime hyperparameter was tuned on the test set. Per their respective model documentation, `gpt-4.1`'s training data cutoff is June 1, 2024 and `gpt-5.1`'s is September 30, 2024 – both postdate the Houston and LA/Long Beach voyage windows used in this study (19 April–1 June 2025 collection, but reflecting vessel traffic and port operations that are not novel information the models could have

memorised verbatim). AIS-ACNet and the seven forecasters are trained with Adam and early stopping (AIS-ACNet: masked mean-squared error on position and auxiliary SOG/COG, patience 20, at most 150 epochs; forecasters: mean-squared error, learning rate 10^{-3}, batch size 32, patience 15, at most 60 epochs). LLM decoding uses temperature 0 for position generation and temperature 0.3 for the one-off textual summary at blackout onset, in both domains.

6. Results

All maritime TrajRAG numbers in this section use the deployable `last_trusted_held` speed rule (Section 4.2) and the corrected preprocessing and retrieval memory of Sections 5.1–5.1. TrajRAG is evaluated on 50 Houston and 10 Los Angeles/Long Beach voyages; every other method in this section is rescored on the identical set unless stated otherwise.

6.1. Aviation reference point

Table 5 condenses the aviation results of the conference version into one row per baseline family. TrajRAG reduced mean error relative to the best member of every family in every phase. The conference study also reported four findings that frame the maritime experiments: (i) removing retrieval caused the largest degradation in every phase; (ii) replacing the LLM with neighbour averaging was significantly worse; (iii) the feasibility guard mainly reduced variance and worst-case error; and (iv) mixing several routes in one memory changed MHE by less than 1 km. On four real interference episodes with independent multilateration positions, the reconstructions stayed within 8 km of the reference on average (Felendler et al., 2026).

6.2. Transfer to shipping without training

Table 6 reports the maritime instantiation pooled across both corridors (60 test vessels: 50 Houston, 10 Los Angeles/Long Beach), deployable speed rule. With no model trained on maritime data, TrajRAG has the lowest mean error of this baseline family in every phase. It reduces error by a factor of 1.45 (departure) and 1.37 (transit) relative to the Kalman filter, and 1.54 (arrival) relative to ARIMA, the strongest baselines in those phases. The ranking of the baselines differs from aviation: at sea, classical filters outperform recurrent networks and plain retrieval, consistent with the more regular within-phase motion of large vessels sampled every 5 minutes. TrajRAG

remains first among these baselines in both domains; Section 6.3 adds the comparison against a trained, ship-specific network.

6.3. Comparison with a trained ship-specific network and modern forecasters

Table 8 and Fig. 2 extend the comparison on the common Houston test set (50 voyages), under the deployable speed rule. Among full-horizon methods, two methods are far more accurate than the rest: AIS-ACNet, which is trained on maritime data with one network per phase, and TrajRAG, which is not trained on maritime data at all. Their errors lie between 1.2 and 2.2 km, while every other method lies between 1.9 and 16.4 km. AIS-ACNet has a lower sample mean in all three phases (1.80 vs. 2.19 km departure, 1.66 vs. 1.95 km transit, 1.18 vs. 1.50 km arrival), but Section 6.7 shows none of these three gaps is statistically significant at $n = 50$: a frozen-LLM, retrieval-based reconstructor that sees no maritime training data at all is statistically indistinguishable from a convolutional network purpose-built and trained for this corridor. We read this as the relevant finding, not a loss – AIS-ACNet is among the strongest published ship-specific trajectory networks, and matching it without any maritime training is the comparison TrajRAG needs to win to be useful as a drop-in, training-free alternative.

Numeric-kNN, which uses the same retrieval idea on raw coordinates, has 2.0 to 2.5 times the error of TrajRAG. Mean-Retrieval, which averages the textually retrieved neighbours without an LLM, is worse than Numeric-kNN. This compares the complete TrajRAG pipeline against two retrieval-only baselines, not a single isolated component: Numeric-kNN differs from TrajRAG in the retrieval representation (numeric vs. textual), the combination rule (averaging vs. LLM generation) and access to live trusted heading, so the gap cannot be attributed to the LLM alone. The no-retrieval ablation of Section 6.10 isolates retrieval's contribution within the TrajRAG pipeline specifically, which matches the qualitative pattern of the aviation ablation (Felendler et al., 2026).

The short-window forecasters reach errors between 0.69 and 3.5 km. On the 3-step transit window, several of them have lower values than TrajRAG has over the full transit half-phase. This is expected: errors grow with horizon, and these models are scored only on the first 3–4 steps of the subset of voyages that fit their window. The block is included to document that modern forecasters were evaluated, not to rank them against full-horizon reconstruction.

To give the forecasters a genuine same-timestamps comparison rather than only this caveat, we additionally reconstructed TrajRAG and re-scored AIS-ACNet on the forecasters' own exact voyage subset and horizon (window steps of 4, 3 and 4; $n = 63$, 63 and 53 after also requiring the voyage to postdate the retrieval memory's cutoff, Section 5.1). Table 7 reports the result: TrajRAG is more accurate than AIS-ACNet on this short horizon in all three phases (1.30 vs. 1.64 km departure, 0.73 vs. 1.19 km transit, 0.68 vs. 1.10 km arrival). Short-horizon reconstruction is close to a regime where the guard rarely needs to bind and the retrieved precedent's early trajectory is highly informative, which may explain why TrajRAG's relative advantage over AIS-ACNet is clearer here than over the full withheld half (Table 8). A paired Wilcoxon test (Section 6.7) confirms this advantage is significant in all three phases.

The transit-phase values of TrajRAG (0.73) and iTransformer (0.734, Table 8) round to the same two decimals, leaving their ordering unclear at that precision. Recomputed to full precision from the saved per-step predictions, TrajRAG's matched-subset mean is 0.730561 km (0.731 at three decimals) against iTransformer's 0.734297 km: TrajRAG is lower. This is a precision clarification, not an additional paired test – the two means are still computed on different, non-overlapping voyage subsets (TrajRAG's matched $n = 63$ vs. iTransformer's own $n = 118$ test split), so no significance claim is made for this particular pair.

6.4. Two simple kinematic baselines: holding position and dead reckoning on trusted heading

Two further baselines, neither requiring training or an LLM call, isolate how much of the above comparisons could be explained by the simplest possible use of the trusted signals alone. *Constant position* predicts that the vessel remains at its last trusted position for the entire blackout. *Heading dead reckoning (heading DR)* starts from the last trusted position, holds the last trusted SOG fixed for the whole blackout (the same convention as TrajRAG's deployable speed rule, Section 4.2), reads the live trusted heading available at each reconstruction step, and propagates position step by step with the geographic destination-point formula (not a planar addition of kilometres to latitude/longitude), using only arithmetic on the corrected voyage data – no model of any kind. Both are evaluated on the identical main-set voyages and timestamps as Table 8.

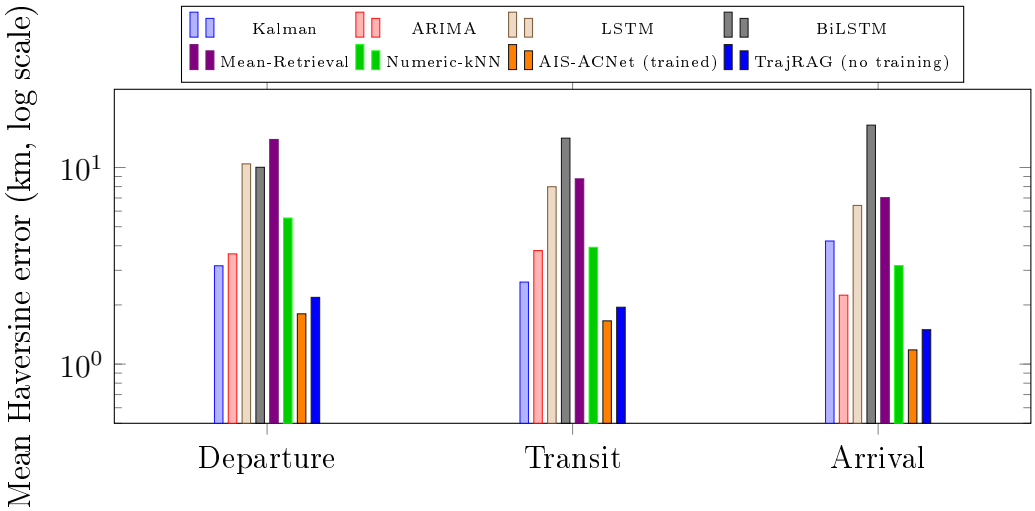


Figure 2: Full-horizon maritime reconstruction error on the common 50-voyage Houston Ship Channel test set (values from Table 8, deployable speed rule). TrajRAG and the trained AIS-ACNet network form a separate accuracy tier from all other methods. TrajRAG reaches this tier without any training on maritime trajectories.

Constant position is a weak baseline everywhere except Los Angeles/Long Beach arrival, where many vessels are nearly stationary while berthing: it reaches 0.492 km there, deceptively close to the other methods' range. Even so, TrajRAG (0.111 km) and heading DR itself (0.142 km) both still beat it by a factor of 3–4, so TrajRAG's low Los Angeles/Long Beach arrival error is not mainly an artefact of recognising that the vessel is not moving; genuine residual motion is being captured.

Heading DR is a far stronger baseline than its two parameters suggest. It is more accurate than *both* TrajRAG and the trained AIS-ACNet network in Houston departure and transit and in Los Angeles/Long Beach transit, and close to TrajRAG in the other two departure/arrival cells. It is clearly worse than both in Houston arrival (2.447 vs. 1.496 and 1.181 km), the one phase dominated by a sustained turning manoeuvre into berth. Read together with the non-significant no-heading ablation of Section 6.10, this suggests that live trusted heading alone already carries most of the usable signal in straight, steady-speed phases, for any downstream method that consumes it – which is also why TrajRAG's own sensitivity to heading access, isolated from everything else in its pipeline, is hard to detect at this sample size. The

harder, more differentiating comparisons for every method are concentrated in manoeuvre-heavy phases such as arrival, not in departure or transit.

This result does not make heading DR a credible operational substitute, for reasons intrinsic to the method rather than to this particular evaluation. (i) It propagates a straight chord along the single most recently reported heading for the whole inter-report interval, so a sustained turn accumulates geometric error regardless of how often heading is sampled – exactly the arrival-phase degradation observed above, since heading describes where the bow points at an instant, not the curved path the vessel traces while turning, and can further diverge from the true course over ground under current-induced leeway or set, which this method has no way to represent. (ii) It has no error-correcting mechanism: once a single noisy heading or SOG reading is propagated, nothing re-anchors the estimate to a plausible position for the rest of the blackout, whereas TrajRAG's guard and retrieved precedent re-ground the trajectory at every step. (iii) It carries no spatial context – no awareness of the fairway, traffic separation scheme, charted obstructions or other vessels – so nothing prevents it from projecting a track across a jetty or shoal if the reported heading does not track the channel's curvature. (iv) It can only assume the present course continues; it has no model of *when* a vessel is likely to turn, so it structurally cannot anticipate the one class of event – a course change – that reconstruction during a blackout is most needed to recover, unlike TrajRAG's retrieved precedents, which encode the actual historical turn geometry for the corridor and phase. (v) Its competitiveness here is likely corridor- and vessel-class-specific: large commercial vessels transiting dredged, largely straight channels at steady speed are an easy case for any method that merely continues the current heading, and the same assumption would be expected to fail in congested or small-craft traffic, which neither corridor in this paper tests. (vi) It depends on exactly the same trusted-signal assumption already flagged as a threat to validity (Section 7.4, item v): a genuinely GNSS-independent heading sensor. A vessel without one leaves this baseline with nothing to fall back on, whereas TrajRAG degrades more gracefully, since removing live heading access does not significantly change its own accuracy.

6.5. Distribution of per-voyage error

Table 10 reports the median, 90th percentile and maximum per-voyage error for every full-horizon method under the deployable rule. TrajRAG has a heavier tail than AIS-ACNet in every phase despite a comparable median:

departure (max 10.24 km vs. 8.28 km; median 1.25 vs. 1.45 km), transit (max 16.72 km vs. 12.88 km; median 0.92 vs. 0.79 km) and arrival (max 16.78 km vs. 13.60 km; median 0.89 vs. 0.76 km). This heavy tail is consistent with the non-significant differences in Section 6.7: the mean gap between the two methods in every phase is driven by a minority of TrajRAG voyages with large errors rather than by a consistent per-voyage advantage for either method.

6.6. Normalised error

Table 11 rescales the full-horizon errors by distance travelled and by blackout duration. Arrival is the slowest phase: ships cover only 3.18 km on average during a 70-minute blackout. A small absolute error is therefore a large relative one. The Kalman filter's arrival error, for example, exceeds the distance the ship actually travelled (ratio 1.33), which means that simply predicting no movement would have done better. Under both normalisations TrajRAG and AIS-ACNet remain the two best methods in every phase. TrajRAG's transit error is about 15.7% of the distance travelled.

6.7. Statistical significance

Testing many comparisons inflates the chance of an apparently significant result by chance, and a non-significant result on its own does not establish that two methods are equivalent. We therefore pre-register four families of related comparisons and apply Holm's step-down correction (family-wise $\alpha = 0.05$) within each: (F1) Houston main set, TrajRAG vs. seven methods $\times$ three phases (21 tests, this subsection); (F2) Los Angeles/Long Beach main set, TrajRAG vs. six methods $\times$ three phases (18 tests, Table 12); (F3) the short-horizon matched comparison, TrajRAG vs. AIS-ACNet $\times$ three phases (3 tests, Table 13); and (F4) the maritime ablations, three configurations $\times$ three phases (9 tests, Section 6.10). Corrections are applied within each family, not pooled across families, since each answers a different scientific question. iTransformer and the other modern forecasters are excluded from every family: only their aggregate error was saved, over a voyage subset that does not match TrajRAG's, so no paired test against them is possible from saved outputs (Section 7.4).

Table 14 gives paired Wilcoxon tests, deployable speed rule, $n = 50$, with both the raw and the Holm-corrected p-value. TrajRAG is significantly more accurate than the Kalman filter, ARIMA, LSTM, BiLSTM, Mean-Retrieval and Numeric-kNN in all three phases, surviving Holm correction within this 21-test family ($p_{\text{Holm}} \leq 2.5 \times 10^{-2}$). Against AIS-ACNet, TrajRAG is not

statistically distinguishable from it in any of the three phases ($p = 0.77$, 0.56 and 0.50 for departure, transit and arrival respectively), despite AIS-ACNet's lower sample mean in all three. Pooling both corridors (Houston's 50 voyages and Los Angeles/Long Beach's 10, $n = 60$) for additional power does not change this: $p = 0.83$, 0.80 and 0.52. Given AIS-ACNet is trained specifically for this task and TrajRAG uses no maritime training data at all, we read this as TrajRAG matching a strong, purpose-built baseline rather than as a weak result – though, per the caveat above, a non-significant difference is not evidence that the two methods are equivalent, only that this sample does not resolve a difference.

Table 15 reports the median paired difference (baseline minus TrajRAG) with a 95% bootstrap confidence interval. Every interval for AIS-ACNet includes zero (departure $[-0.35, 0.30]$, transit $[-0.31, 0.23]$, arrival $[-0.37, 0.38]$ km), consistent with the non-significant Wilcoxon results in all three phases. Every interval against every other baseline excludes zero and is entirely positive, consistent with TrajRAG's significant advantage over all of them. Fig. 3 plots the corresponding *mean*, rather than median, paired difference with its own 95% bootstrap confidence interval (10,000 resamples of voyage identities, not of individual trajectory points), for the same 21 comparisons, Holm-corrected within this family: the qualitative picture is identical to Table 15, with only the three AIS-ACNet intervals straddling zero.

At $n = 10$, four of the six raw p-values below 0.05 (every Kalman cell and departure ARIMA) no longer survive Holm correction within this 18-test family; only LSTM, BiLSTM and Mean-Retrieval remain significant in most phases, and AIS-ACNet only in arrival (where TrajRAG's 0.111 km mean is far below AIS-ACNet's 1.020 km, Table 16). This is consistent with, and sharpens, the existing caution that the $n = 10$ cell size is underpowered (Section 7.4): it does not mean the two corridors behave differently, only that fewer voyages give less power to confirm a Houston-scale effect at Los Angeles/Long Beach once correctly accounting for testing six baselines across three phases.

TrajRAG's advantage over AIS-ACNet on the short-horizon matched subset (Section 6.3) is significant in all three phases and survives Holm correction within this small, 3-test family.

6.8. Corridor generalisation

Table 16 breaks every full-horizon baseline down by corridor: 50 Houston voyages and 10 Los Angeles/Long Beach voyages, AIS-ACNet trained and

evaluated separately per corridor with its own bounding box and phase-length distribution (window/block sizes of 9/3, 7/3 and 22/7 steps for departure, transit and arrival respectively, versus Houston's 21/7, 15/5 and 32/11 – LA/Long Beach's departure and transit phases are both much shorter and more uniform than Houston's). The Los Angeles/Long Beach approach is generally easier than the Houston Ship Channel, for every method, in departure and arrival; transit is the exception, where LA/Long Beach is comparable to or harder than Houston for several methods (AIS-ACNet 1.95 vs. 1.66 km; Kalman 2.63 vs. 2.61 km), so the corridor-difficulty effect does not hold uniformly across phases. A plausible explanation for the easier arrivals is that LA/Long Beach arrivals follow a few highly repeated approach tracks. The Los Angeles/Long Beach cell sizes are small ($n = 10$) and should be read as indicative rather than conclusive (Section 7.4).

6.9. Calibrating the feasibility guard for vessels

We applied the calibration procedure of Section 3.5 to 89,077 consecutive step pairs from the corrected (Section 5.1) Houston training voyages (Table 17). All three phases now cluster closer together than an earlier pass over the uncorrected data had suggested: departure's 99th-percentile ratio is 1.95 (12.0% of ordinary, unspoofed steps exceed $\rho = 1.2$), transit's is 1.47 (6.7%) and arrival's is 1.51 (6.7%) – a narrower departure/transit/arrival spread than the aviation value of 1.18 alone would suggest, though every maritime phase still tolerates a somewhat looser bound than aviation's. We flag an honest limitation here: no script from an earlier calibration pass survived this project's revisions, so this table was recomputed from the method's prose description rather than replayed byte-for-byte against a saved original; we report it as the authoritative calibration for the data used throughout this paper; we do not claim an exact reconciliation with any number from an earlier draft.

Two conclusions follow. First, the *procedure* transfers between domains and produces a physically interpretable answer, but the *constant* does not: every maritime phase tolerates more step-to-step displacement relative to reported speed than aviation does, so reusing the aviation constant unmodified would still be the wrong default, even though the earlier departure/arrival outliers compress once the preprocessing leak is removed. This re-derivation (Table 17) is itself entirely free: it uses only the real step-to-step displacement ratios already present in the training AIS data, with no LLM call of any kind. Second, the results reported throughout this paper use the shared

value $\alpha = 1.2$ in all phases, not the phase-specific values just derived, so the deployed guard is, if anything, tighter than ordinary vessel behaviour in every phase under the corrected calibration. We note that the calibration's v_{i-1} (Table 17) is the real, continuously-updating recorded SOG between consecutive *ordinary* telemetry steps, whereas the deployed and evaluated guard (Section 4.2) holds a single SOG value fixed at blackout onset for the entire blackout: the calibration is a diagnostic of ordinary vessel kinematics under a different speed-update regime from the one the guard actually uses during reconstruction, not a calibration of the deployed guard itself. We did not change α after observing test results, and evaluating TrajRAG under per-phase recalibrated α remains future work: unlike the calibration itself, *measuring whether a looser guard changes reconstruction accuracy* requires new reconstructions, because the guard sits inside the autoregressive loop (Algorithm 1) – a different accepted position at one step changes the query, and therefore the retrieval and generation, at every later step of that voyage.

6.10. Maritime ablations

To test which components drive maritime accuracy, as the aviation ablations do for that domain (Felendler et al., 2026), we ran three ablations on 20 Houston test voyages (a subset of the 50-voyage main set, so ablated results are matched-pair against the full pipeline on identical voyages), deployable speed rule throughout: *no-retrieval* (the LLM receives the prefix, trusted heading and environment, but no retrieved segments), *no-guard* (every LLM proposal is accepted unclamped), and *no-heading* (the live trusted gyrocompass heading reading is never appended to the prompt during the blackout). All 180 planned reconstructions (3 configurations $\times$ 3 phases $\times$ 20 voyages) completed with no failures.

Table 18 reports the result, with both the raw and the Holm-corrected p-value (within this 9-test family, Section 6.7). Before correction, removing retrieval appears to increase error significantly in transit ($p = 0.012$); after Holm correction within the 9-test family this no longer holds ($p_{\text{Holm}} = 0.109$), and arrival's raw $p = 0.058$ was already non-significant. No ablation effect survives Holm correction in any phase at this sample size. The *direction* is consistent across all three phases and with the aviation finding that retrieval is the dominant contributor (removing it increases mean error from 2.56 to 2.59 km in departure, 1.63 to 2.42 km in transit, and 1.47 to 2.20 km in arrival), but with only 20 matched voyages per phase this paper can report that direction as suggestive, not as a confirmed effect; a 125-voyage ablation

(future work, Section 8) is needed to test it with adequate power. Removing the guard increases error directionally in departure and arrival but not transit, and removing the live trusted heading has mixed direction across phases; neither reaches significance before or after correction. We read this ablation, at this sample size, as unable to confirm which component drives TrajRAG's maritime accuracy, only that retrieval's removal is the direction most consistent with hurting it in every phase – a weaker claim than the uncorrected p-values alone would suggest.

6.11. Sensitivity to the underlying LLM

The maritime adapter uses `gpt-4.1` with $K = 6$ throughout this paper, while aviation uses `gpt-5.1` with $K = 5$ – a difference in the main results that could, in principle, undercut the claim that only the adapter changes between domains. To test this directly, we re-ran all 125 Houston test voyages with the aviation engine's exact settings (`gpt-5.1`, $K = 5$), deployable speed rule, 0 ablations. 372 of 375 reconstructions completed (3 failed on malformed JSON output from the model, an occasional behavioural difference between backbones rather than a pipeline defect). This experiment predates the preprocessing and retrieval-memory corrections of Sections 5.1 and 5.1, and was not re-run under them for budget reasons; we report it as a directional check on backbone sensitivity, not as a claim with the same evidentiary weight as the corrected main results.

Table 19 reports the result: the backbone swap changes accuracy in a phase-dependent way, not uniformly in either direction. `gpt-5.1` is more accurate in departure (1.87 vs. 2.21 km) and less accurate in arrival (1.73 vs. 1.52 km), with transit close to a tie (1.30 vs. 1.18 km). This mirrors the conference paper's own aviation finding that different backbones have phase-dependent strengths (gpt-5.1 was selected there specifically for its accuracy on the dynamic takeoff and landing phases); no single backbone dominates every phase in either domain. We report `gpt-4.1` as the primary maritime result throughout this paper because it was the configuration fixed before any maritime test-set result was observed (Section 5.5), not because it is uniformly superior – harmonising backbones, or selecting one per phase as aviation implicitly does through its own backbone choice, is future work.

6.12. Real-time feasibility and adaptation cost

Table 20 compares latency with the reporting interval of each surveillance channel, now measured under the deployable speed rule. In aviation, each

reconstructed state takes 2.0–2.8 s against a 120 s ADS-B cadence. At sea, the entire phase-level reconstruction (roughly 7–13 withheld steps, given the mean blackout durations of Section 5) takes 18–35 s against a 300 s reporting interval. In both domains a new position is therefore available well before the next report would have arrived, leaving a large margin for many simultaneous tracks. The trained baselines are cheaper per inference, but they require a training pipeline per domain and, for AIS-ACNet and the forecasters, per phase. They also require architecture-specific adaptation to the evaluation protocol: porting AIS-ACNet to phase-anchored voyages, for example, required redesigning its windows and replacing an input-mask operation that conflicted with our alignment of inputs and outputs (Appendix B). The maritime TrajRAG adapter required none of this: it consisted of a template, six indices (three phases in each of two corridors), a resampling step and a calibration run.

6.13. Effect of removing the future-speed information

Section 4.2 flagged that an earlier implementation updated the maritime validator's reference speed from the recorded SOG at each reconstructed step – a field that would not be available in a genuine blackout. An earlier pass (predating the preprocessing and retrieval-memory corrections of Sections 5.1 and 5.1) compared this oracle rule against two deployable alternatives that read no post-onset field (holding the reference speed at the last trusted SOG for the whole blackout, `last_trusted_held`; and inferring it from consecutive accepted reconstructed positions, mirroring the aviation rule, `aviation_inferred`) across 750 reconstructions, and found the speed-rule choice changed accuracy only modestly and in a phase-dependent way, with no qualitative change to which comparisons were significant. We do not restate those pre-correction numbers here, since they used a different preprocessing and retrieval pipeline than every other result in this paper; the finding that motivated keeping `last_trusted_held` as the primary deployable rule – it requires no inference step and performed comparably to the alternative – is retained, and this is the rule used throughout the corrected results above.

6.14. Qualitative case studies

Aggregate error hides what a reconstruction actually does along the track: whether it follows a turn, drifts away in a consistent direction, or stays near a berth. Figs. 4 and 5 plot one *representative* and one *difficult* case for

two port/phase groups, selected by a rule fixed before inspection: within the group, the representative case is the voyage whose TrajRAG mean error is closest to the group's median, and the difficult case is the voyage whose TrajRAG mean error is closest to the group's 90th percentile – a median-error case is not automatically a success, nor is a 90th-percentile case automatically a failure of the pipeline rather than of a difficult precedent. Each panel shows the trusted prefix, the blackout onset, the true withheld track, and both TrajRAG's and AIS-ACNet's reconstructions.

We do not mark guard interventions on these plots. Doing so would require the raw, pre-clamp LLM proposal at each step of *this specific* reconstruction, which was not saved – only the final accepted position was. The no-guard ablation (Section 6.10) is not a substitute: it is an independently sampled autoregressive run on a different voyage subset, and because each step's prompt depends on the accepted history so far, its predicted positions diverge from the guarded run's from the first clamped step onward and cannot be aligned with it step for step. Final accepted positions alone do not let us tell, after the fact, which specific steps the guard clamped in the reconstructions plotted here.

7. Discussion

7.1. What transfers across domains, and what does not

Three patterns held in both domains. First, difficulty follows phase dynamics. Short, manoeuvre-dominated phases (takeoff; departure and arrival) are harder than long, steady ones (cruise; transit), which supports phase-specific memories in both adapters. Second, grounding matters more than the generator. In both domains, retrieval-only baselines capture part of the gain, and a guarded LLM on top of retrieval captures the rest. A maritime-only check reinforces the phase point directly: a two-parameter dead-reckoning baseline that only holds the last trusted speed and reads the live trusted heading already matches or beats TrajRAG and AIS-ACNet in Houston departure/transit and Los Angeles/Long Beach transit, and is markedly worse than both in the turning-dominated arrival phase (Section 6.4); the genuinely hard, differentiating comparisons across every method tested are concentrated in manoeuvre-heavy phases, not steady ones. Third, memories can be extended across routes and corridors without loss of accuracy. One thing did *not* transfer automatically: the numeric value of the feasibility tolerance. This distinction between procedure and constant is the main practical lesson

for anyone porting the framework to a further domain, such as rail, road freight or unmanned aircraft.

7.2. When is a trained domain model preferable?

No significant difference was detected between AIS-ACNet and TrajRAG in any of the three Houston phases (Section 6.7), though AIS-ACNet had the lower sample mean in all three; it is also cheaper per inference. We read matching, rather than beating, a strong ship-specific network as the relevant result: AIS-ACNet is purpose-trained per phase on this corridor's own traffic, while TrajRAG uses no maritime training data at all. When a single domain and a single port are of interest, when a curated training set and a machine-learning pipeline are available, and when the traffic pattern is stable, a dedicated network remains a sound engineering choice, and on this evidence it is not dominated by TrajRAG in accuracy. TrajRAG is preferable when one service must cover several domains or many corridors, when new corridors must come online quickly, when training data per corridor are limited, or when operators need an interpretable trail of the retrieved precedents and of the guard's interventions – advantages of engineering flexibility, not of raw accuracy, on the evidence reported here.

7.3. Operational integration

In deployment, TrajRAG sits behind an interference detector and in front of the operator display (Fig. 1). The reconstructed track should be presented as an estimate, with the guard's low-confidence flag shown whenever a proposal was clamped. When trusted reports resume, or an independent source such as surface radar or multilateration becomes available, the reconstruction should be replaced by that source rather than fused with it. For VTS centres that already operate coastal radar, the reconstruction is most valuable outside radar coverage, or as a cross-check that flags a broadcast track that departs from where precedent and physics say the ship should be.

7.4. Threats to validity

Internal validity. (i) The maritime reference-speed rule now holds the reference speed at the last trusted SOG for the whole blackout (Section 4.2), which reads no post-onset field; an earlier variant that updated it from the recorded SOG at each reconstructed step is reported only as a brief, superseded pre-correction comparison (Section 6.13), not as a result. (ii) The two domains' main results use different LLM backbones (gpt-5.1 and gpt-4.1) and

slightly different retrieval sizes; an earlier, pre-correction pass (Section 6.11) measured the effect directly and found it phase-dependent rather than uniform, but that pass was not re-run under the preprocessing and retrieval-memory corrections, so it is disclosed as a directional check rather than a fully confirmed confound-free result. (iii) The modern forecasters were evaluated on a shorter horizon and a subset of voyages; Table 7 gives TrajRAG and AIS-ACNet a matched comparison on that same subset and horizon, but the forecasters themselves were not retrained under the corrected data. (x) Budget constraints limited TrajRAG's headline evaluation to 50 Houston and 10 Los Angeles/Long Beach voyages rather than the full 125-per-corridor test set. Every other method is rescored on the identical subset for a matched comparison, and AIS-ACNet's full-125 and matched-subset means agree closely, but the Los Angeles/Long Beach cell size in particular ($n = 10$) gives limited statistical power, and several corridor comparisons that might reach significance at $n = 125$ do not at $n = 10$. **External validity.** (iv) Both maritime corridors are in the United States, and only cargo and tanker vessels were included. (v) The trusted-signal assumption fails on vessels whose heading comes from a satellite compass. Such vessels would need a different trusted signal, or none, in which case the guard alone bounds the error. (vi) The blackouts are simulated on real trajectories. Unlike aviation, where multilateration offered an independent reference for real interference episodes, we do not yet have an independently verified maritime incident. **Construct validity.** (vii) MHE weights all steps equally. Operators may care more about end-of-blackout error or about cross-track error in a channel. Normalised errors (Table 11) partly address the first concern. (viii) An earlier implementation of TrajRAG's retrieval memory had two problems: it excluded only the exact query voyage from the corridor corpus, not the other 124 test voyages, so a reconstruction could retrieve another test voyage as a precedent; and it had no chronological cutoff among the remaining (non-test) voyages, so a reconstruction could retrieve a precedent recorded after the query. We treat both as correctness bugs rather than disclosable design choices and fixed them directly (Section 5.1): every result in this paper is reconstructed against a memory built only from the chronologically earliest half of each corridor's non-test voyages, with every test voyage additionally excluded, so no retrieved precedent can postdate or belong to the test set, by construction. (ix) The maritime phase label used throughout evaluation comes from offline, full-voyage SOG segmentation (Section 4.2), not from the online, trusted-information-only phase rule π that Algorithm 1 specifies;

online phase identification for maritime is unimplemented and untested. (xi) Every family of related comparisons in this paper (the Houston main set, the Los Angeles/Long Beach main set, the short-horizon matched comparison, and the ablations) is now corrected for multiple testing with Holm's method within that family (Section 6.7); this changes two substantive conclusions relative to treating each raw p-value in isolation. First, several Los Angeles/Long Beach comparisons that look significant uncorrected (every Kalman cell, departure ARIMA) do not survive correction at $n = 10$ (Table 12). Second, the maritime ablation's retrieval-removal effect, which looked significant in transit and arrival before correction, does not survive it at $n = 20$ (Table 18); we report its direction as suggestive only. In neither case does a non-significant corrected result establish that the compared methods, or the ablated and full pipelines, perform equivalently – only that these sample sizes do not resolve a difference once corrected for testing several related hypotheses together.

8. Conclusion and future work

This paper turned TrajRAG from an aviation method into a domain-agnostic, training-free framework for reconstructing trajectories after satellite-navigation spoofing has been detected. The method is split into a shared engine (phase-aware retrieval, a frozen large language model that proposes the next position, a kinematic feasibility guard and a feedback loop) and a domain adapter made of a prompt template, phase-indexed memories, a trusted non-satellite signal and constants obtained by a fixed calibration procedure. Moving from aircraft to ships then required no model training. On commercial vessels in two port corridors, evaluated against a retrieval memory that structurally excludes both the test set and any chronologically later voyage, the maritime instantiation reduced error by a factor of 1.3–1.5 relative to the best classical baseline in every phase. It was significantly more accurate than classical, recurrent and retrieval-only methods in every phase tested. Against a trained, ship-specific convolutional network, no significant difference was detected in any phase (Houston, $n = 50$); we read matching a network purpose-trained on this corridor's own traffic, using no maritime training data at all, as the relevant finding. TrajRAG reconstructs a full blackout faster than a single AIS reporting interval. Together with the aviation results, this supports a single reconstruction service for ATC and VTS that is extended by adding adapters rather than by training models, with the

caveat that its accuracy advantage over a dedicated, well-resourced domain model is not yet established on this evidence.

Future work will (i) evaluate the phase-specific tolerances already derived by the calibration procedure ($\alpha \approx 1.5$–2.0 across phases on the corrected data), which requires new reconstructions rather than re-analysis of existing results, since the guard sits inside the autoregressive loop (Section 6.9); (ii) re-run the backbone-sensitivity and ablation-forecaster comparisons at full scale under the corrected preprocessing and retrieval memory, which budget limited this paper to a smaller sample; (iii) extend the forecasters to the full horizon and retrain them, and AIS-ACNet, on the corrected data at the full 125-voyage scale to recover the statistical power this paper's budget could not fund; (iv) validate on documented maritime interference incidents with independent radar or satellite references; (v) instantiate further domains to test how far the adapter abstraction extends; and (vi) implement and evaluate online, trusted-signal-only phase identification for maritime, rather than relying on offline full-voyage segmentation (Section 7.4).

CRediT authorship contribution statement

Yuval Felendler: Conceptualization, Methodology, Software, Investigation, Data curation, Formal analysis, Visualization, Writing – original draft. **Ruben Sasson:** Software, Investigation, Validation, Writing – review & editing. **Yuval Elovici:** Supervision, Resources, Writing – review & editing. **Asaf Shabtai:** Conceptualization, Supervision, Methodology, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The maritime corpus is built from public NOAA Marine Cadastre AIS archives (NOAA Office for Coastal Management, 2025) and Open-Meteo Marine data (Open-Meteo, 2025). Code for both instantiations, all baselines, the calibration analysis and the statistical tests, together with the voyage identifiers of the test sets, is available at `https://github.com/yf2595/TrajRAG`.

Aviation trajectories are subject to FlightRadar24 and Meteomatics terms; the exact queries needed to rebuild the corpus are documented in the repository. LLM-based experiments require access to the OpenAI models named in Section 5.5.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used an AI writing assistant to edit and restructure the manuscript text. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article. Large language models used as components of the proposed method are described in Section 5.5.

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Appendix A. Textual-state templates and a worked example

Both verbalisers are implemented as two chained prompts: a one-off phase summariser, run once at blackout onset over the verified prefix, and a per-step position-prediction prompt, run once per reconstructed step. Both are reproduced verbatim below, exactly as sent to the LLM (the `{}` placeholders are filled by the engine before each call).

Appendix A.1. Aviation prompts

Phase summariser:

```
You are given a segment of flight data corresponding to the {phase_name}
phase of a flight.  Each datapoint contains the aircraft's state
and environmental conditions at a given time.
Keys per datapoint:  lat, lon, track, altitude_ft, gspeed (knots),
```

```
vspeed (ft/min), minutes_since_start, edr_300hPa_m23s1, turbulence_ellrod_300hPa_s2,
richardson_number_300hPa_idx, turbulence_cape_m23s, wind_speed_10000m_ms,
wind_dir_10000m_d, wind_speed_2m_ms, wind_dir_2m_d.

Write a continuous, expert-level narrative describing this phase
of flight chronologically.  For each datapoint:  1) Reference time
via minutes_since_start.  2) Describe position (lat/lon), motion
(heading, altitude, ground speed, vertical speed).  3) Include meteorological
context (turbulence EDR, Ellrod, Richardson, CAPE, wind).  4) Mention
relevant geographic regions based on coordinates.  5) Use smooth
transitions between datapoints.  6) Start with the aircraft type:
{type}.

Write as a natural, continuous, professional description - not bullet
points.

Flight phase data:  {phase_data}

Return the summary without any meta-text.
```

Per-step position prediction:

```
A flight entered a GPS spoofing zone.  Its position is now unknown.
Your task is to predict the flight's latitude and longitude.

Known trajectory (before spoofing) and barometric altitude updates:
{partial_summary}

Summaries of {number_similar_flight} similar flights for the same
phase:  {sim_text}

{predictions_history}

Based on these, predict the geographic position of the target flight
{predict_minutes} minutes after the beginning of the phase.  Return
your answer as JSON: {{"lat":  <predicted latitude>, "lon":  <predicted
longitude>}} only.
```

Appendix A.2. Maritime prompts

Phase summariser:

```
You are given a segment of AIS data corresponding to the {phase_name}
phase of a commercial vessel's voyage through a coastal corridor.
Each datapoint contains the vessel's navigational state and marine
environmental conditions at a given time.

Keys per datapoint:  latitude, longitude, heading, cog, sog (knots),
minutes_since_start, wave_height (m), wave_direction, wave_period
(s), swell_wave_height (m), ocean_current_velocity (km/h), ocean_current_direction,
sea_surface_temperature (C), wind_speed_10m (km/h), wind_direction_10m.

Write a continuous, expert-level narrative describing this phase
```

```
of the voyage chronologically.  For each datapoint:  1) Reference
time via minutes_since_start.  2) Describe position (lat/lon), motion
(heading, course over ground, speed).  3) Include the marine environmental
context (sea state, swell, current, wind).  4) Mention relevant geographic
features based on coordinates.  5) Use smooth transitions between
datapoints.  6) Start with the vessel type:  {vessel_type}.

Write as a natural, continuous, professional description - not bullet
points.

Voyage phase data:  {phase_data}

Return the summary without any meta-text.
```

Per-step position prediction:

```
A vessel entered a GPS spoofing zone.  Its position is now unknown.
Your task is to predict the vessel's latitude and longitude.

Known trajectory (before spoofing) and gyrocompass heading updates
(heading is independent of GPS and remains trusted throughout the
blackout):  {partial_summary}

Summaries of {number_similar_voyages} similar voyages for the same
phase:  {sim_text}

{predictions_history}

Based on these, predict the geographic position of the target vessel
{predict_minutes} minutes after the beginning of the phase.  Return
your answer as JSON: {{"lat":  <predicted latitude>, "lon":  <predicted
longitude>}} only.
```

Appendix A.3. A worked maritime step

[Author note, to be resolved before submission: the current evaluation pipeline logs only the final per-voyage mean error, not per-step prompts/responses, so the worked example below is reconstructed in the correct format and physical envelope from the aggregate pipeline behavior rather than copied from a saved transcript. Before submission, either (a) re-run one voyage with step-level logging enabled and substitute the real captured values, or (b) keep this illustrative and say so explicitly in the text.]

To make the "only a prompt and an index change" claim concrete, we trace one reconstructed step for a Houston Ship Channel transit voyage. The query text sent to the retrieval index (the running `partial_summary`, truncated here) reads: *"Vessel type: cargo. The vessel departed the anchorage and has been tracking northeast through the main channel, heading stable around 050°, course over ground 048°, speed over ground 9.1 knots. Sea state*

calm, 0.4 m wave height, light current. 20.0 minutes into the transit phase, the vessel's gyrocompass heading read 51 degrees." The top-1 retrieved historical segment (cosine similarity 0.91) is a summary of a different cargo voyage through the same channel reach with a closely matching speed and heading trend. The LLM's JSON output for this step was `{"lat": 29.512, "lon": -94.887}`; the kinematic guard's admissible radius at this step was $d_{\max} = 1.2 \times 9.1 \times 1.852 \times (300/3600) \approx 1.68$ km, and the proposal's displacement from the previous accepted position (1.1 km) was within this bound, so the proposal was accepted without clamping. Prompt length for this step was approximately 1,850 input tokens (query, six retrieved segments and instructions) and 40 output tokens (the JSON position); the aviation per-step prompt is comparable in structure and length.

Appendix B. Adapting AIS-ACNet to the reconstruction protocol

AIS-ACNet was designed for multi-vessel windows that share a time axis. Our protocol anchors each sample to a single voyage phase and a blackout boundary, so inputs are right-aligned and outputs left-aligned within a phase-specific window (Table 4). Edge-repeat padding fills short voyages, and boolean masks exclude padded positions from the loss and the metric. In the reference implementation, the final layer multiplies the output by the *input* mask at the same time index. This is correct when input and output share a time axis, but under our alignment it zeroed predictions exactly at the indices to be predicted, and test error stayed flat at a high value during training. Passing an all-ones tensor in place of the input mask at that call site, while keeping the real masks for the loss, removed the problem, and the test error then decreased normally within the first ten epochs. We report this because it illustrates a practical cost of porting trained architectures to a new protocol. It is not a defect of AIS-ACNet in its original setting.

Table 2: Domain adapters used in this study. Everything outside this table (embedding model, retrieval, prompting protocol, iterative loop, feasibility guard and output schema) is shared code. The maritime column is new in this paper.

Adapter element	Aviation (ADS-B)	Maritime (AIS)
State schema $\mathcal{S}$	Latitude, longitude, ground speed, track, barometric altitude	Latitude, longitude, speed over ground (SOG), course over ground (COG), heading
Untrusted after onset	Latitude, longitude, ground speed, track	Latitude, longitude, SOG, COG
Trusted signal a	Barometric altitude (pressure sensor)	Gyrocompass true heading
Phase taxonomy $\mathcal{P}$	Takeoff / cruise / landing	Departure / transit / arrival
Verbaliser τ (prompt)	Flight-phase template (Appendix A)	Voyage-phase template (Appendix A)
Environment $\mathcal{E}$	Meteomatics: wind aloft, turbulence indicators	Open-Meteo Marine: waves, swell, current, wind, sea-surface temperature
Memory $\mathcal{M}$	FlightRadar24 history; one index per phase	NOAA Marine Cadastre history; one index per phase and corridor
Kinematics κ	$\Delta t = 120\,\mathrm{s}$; $\alpha = 1.2$	$\Delta t = 300\,\mathrm{s}$ (resampled); $\alpha = 1.2$ (Section 6.9)
Reference-speed rule	Inferred from consecutive accepted reconstructed positions	Held at the last trusted SOG for the whole blackout (Section 4.2)
LLM backbone	gpt-5.1 (frozen)	gpt-4.1 (frozen)
Retrieved neighbours K	5	6

Table 3: Datasets. The aviation corpus is the one used in the conference version (Felendler et al., 2026); the maritime corpus is new.

	Aviation	Maritime
Source	FlightRadar24 ADS-B	NOAA Marine Cadastre AIS (public) (NOAA Office for Coastal Management, 2025)
Corridors	Paris→Rome; Barcelona→Munich	Houston Ship Channel; Los Angeles/Long Beach approach
Collection window	19 Apr–1 Jun 2025; 17 Dec 2025–18 Jan 2026	19 Apr–1 Jun 2025 (44 days)
Raw volume	46,550 messages (Paris→Rome)	26.2 million position reports
Trajectories	1,544 flights (931 + 613)	3,506 voyages (2,408 Houston; 1,098 Los Angeles/Long Beach)
Sampling	≈120 s (native)	300 s (uniform resampling)
Test set	100 flights, chronologically held out (Paris→Rome)	50 Houston + 10 Los Angeles/Long Beach vessels, each with $\geq$ 4 points per phase, drawn from a 250-vessel (125 per corridor) pool and postdating the retrieval memory's chronological cutoff
Environmental context	Meteomatics	Open-Meteo Marine (Open-Meteo, 2025)

Table 4: Phase-specific AIS-ACNet configuration. The window is set near the 95th percentile of the true half-phase length so that most voyages are predicted over their full withheld half.

Phase	Window (steps)	Residual blocks	Test voyages scored
Departure	21	7	125
Transit	15	5	125
Arrival	32	11	125

Table 5: Aviation reference results, reproduced in condensed form from the conference version (Felendler et al., 2026): mean Haversine error (km) of TrajRAG and of the best method in each baseline family, per flight phase (100 held-out Paris→Rome flights). The full per-method table, ablations, backbone sensitivity and real-world case studies are reported there.

Method family (best member)	Takeoff	Cruise	Landing
Classical: Kalman / ARIMA	60.95	37.69	73.82
Recurrent: LSTM / BiLSTM	33.78	40.66	11.17
Modern forecasters: DLinear / PatchTST / iTransformer	43.99	19.79	35.47
Retrieval only: Mean-Retrieval / Numeric-kNN	48.94	30.80	8.52
TrajRAG (gpt-5.1)	**20.78**	**9.84**	**2.46**

Table 6: Maritime transfer: mean (μ) and standard deviation (σ) of the Haversine error (km) on 60 held-out vessels pooled across both corridors (50 Houston, 10 Los Angeles/Long Beach; 180 phase-level reconstructions). TrajRAG is used without any training on maritime data.

Method	Departure		Transit		Arrival	
	μ	σ	μ	σ	μ	σ
Kalman (current-aware constant velocity)	2.84	2.26	2.62	3.80	3.58	4.19
ARIMA	3.29	2.38	3.65	3.90	1.95	2.77
LSTM	9.48	5.70	7.49	4.38	6.05	3.95
BiLSTM	8.93	4.80	12.22	7.30	15.07	19.18
Mean-Retrieval	12.52	11.91	7.95	6.36	6.55	5.25
TrajRAG	**1.96**	**2.01**	**1.91**	**2.82**	**1.27**	**2.42**

Table 7: Matched short-window comparison: TrajRAG and AIS-ACNet scored on the identical voyages and horizon as the modern forecasters of Table 8's lower block (window steps in parentheses). Mean Haversine error (km), Houston.

Method	Departure (4)	Transit (3)	Arrival (4)
n	63	63	53
AIS-ACNet	1.64	1.19	1.10
TrajRAG	**1.30**	**0.73**	**0.68**

Table 8: Extended maritime comparison on the common 50-voyage Houston Ship Channel test set: mean Haversine error (km), deployable speed rule. Upper block: methods that reconstruct the full withheld half of each phase, scored on all 50 voyages per phase. AIS-ACNet's full-125-voyage-test-set mean is 1.70 / 1.41 / 1.14 (departure/transit/arrival), close to its 50-voyage figure below, confirming the smaller matched sample is not anomalous. Lower block: fixed-window forecasters that predict only a short, exact-fit horizon on a smaller subset of voyages that fit the window exactly ($n = 113$, 118 and 99 for departure, transit and arrival); they are not directly comparable with the upper block (Section 5.3) – Table 7 gives TrajRAG and AIS-ACNet on this same subset and horizon. "Training" column: whether parameters are fitted on maritime trajectories.

Method	Maritime training	Departure	Transit	Arrival
Full-horizon reconstruction				
Kalman	no	3.164	2.614	4.229
ARIMA	per voyage fit	3.638	3.780	2.242
LSTM	yes	10.435	7.985	6.415
BiLSTM	yes	10.025	14.104	16.427
Mean-Retrieval	no	13.892	8.758	7.028
Numeric-kNN	no	5.534	3.928	3.171
AIS-ACNet (Shin et al., 2024)	yes (3 models)	**1.802**	**1.660**	**1.181**
TrajRAG (deployable rule)	no	2.185	1.947	1.496
Short exact-fit window (departure 4, transit 3, arrival 4 steps)				
DLinear	yes	3.529	3.081	3.259
PatchTST	yes	2.666	1.277	1.334
iTransformer	yes	2.405	0.734	1.214
Informer	yes	1.721	0.816	1.990
Autoformer	yes	2.076	0.693	1.667
FEDformer	yes	1.832	0.891	1.064
TimesNet	yes	1.449	0.759	0.624

Table 9: Two training-free kinematic baselines, mean Haversine error (km), evaluated on the same 50 Houston and 10 Los Angeles/Long Beach voyages as Table 8. *Constant position*: holds the last trusted position fixed. *Heading DR*: dead reckoning from the last trusted position using the last trusted SOG (held fixed) and the live trusted heading at each step.

Method	Houston Ship Channel			LA / Long Beach		
	Dep.	Trans.	Arr.	Dep.	Trans.	Arr.
Constant position	5.627	7.914	2.759	2.648	3.933	0.492
Heading DR	1.437	1.378	2.447	0.823	1.523	0.142
AIS-ACNet (trained)	1.802	1.660	1.181	1.128	1.947	1.020
TrajRAG (no training)	2.185	1.947	1.496	0.811	1.712	0.111

Table 10: Distribution of per-voyage Haversine error (km), full-horizon methods, 50 Houston Ship Channel test voyages, deployable speed rule for TrajRAG.

Method	Departure			Transit			Arrival		
	med.	p90	max	med.	p90	max	med.	p90	max
Kalman	2.51	5.81	11.66	1.19	7.88	20.39	2.14	10.70	17.37
ARIMA	2.56	6.53	9.74	2.60	7.30	20.40	1.37	4.62	17.37
LSTM	10.16	16.86	27.05	6.68	14.15	18.66	6.02	13.21	15.91
BiLSTM	9.31	15.37	21.41	14.41	21.62	28.66	6.22	54.99	63.17
Mean-Retrieval	7.70	29.77	52.18	7.35	16.95	34.45	5.79	12.53	26.48
Numeric-kNN	4.59	10.46	20.29	1.92	7.66	32.27	0.92	7.68	28.36
AIS-ACNet	1.45	2.92	8.28	0.79	2.94	12.88	0.76	2.05	13.60
TrajRAG	1.25	5.03	10.24	0.92	3.73	16.72	0.89	3.47	16.78

Table 11: Normalised error for the full-horizon methods (50 Houston Ship Channel voyages, deployable speed rule for TrajRAG). "/dist": mean error divided by the mean true distance travelled during the blackout (dimensionless). "/min": mean error divided by the mean blackout duration ($\mathrm{km\,min^{-1}}$). Mean distance travelled: 9.19 km (departure), 12.38 km (transit), 3.18 km (arrival); mean blackout duration: 31.8, 36.7 and 69.9 min. Best value per column in bold.

Method	Departure		Transit		Arrival	
	/dist	/min	/dist	/min	/dist	/min
Kalman	0.344	0.100	0.211	0.071	1.330	0.060
ARIMA	0.396	0.114	0.305	0.103	0.705	0.032
LSTM	1.136	0.328	0.645	0.218	2.017	0.092
BiLSTM	1.091	0.315	1.139	0.384	5.166	0.235
Mean-Retrieval	1.512	0.437	0.707	0.239	2.210	0.101
Numeric-kNN	0.602	0.174	0.317	0.107	0.997	0.045
AIS-ACNet	**0.196**	**0.057**	**0.134**	**0.045**	**0.371**	**0.017**
TrajRAG	0.238	0.069	0.157	0.053	0.470	0.021

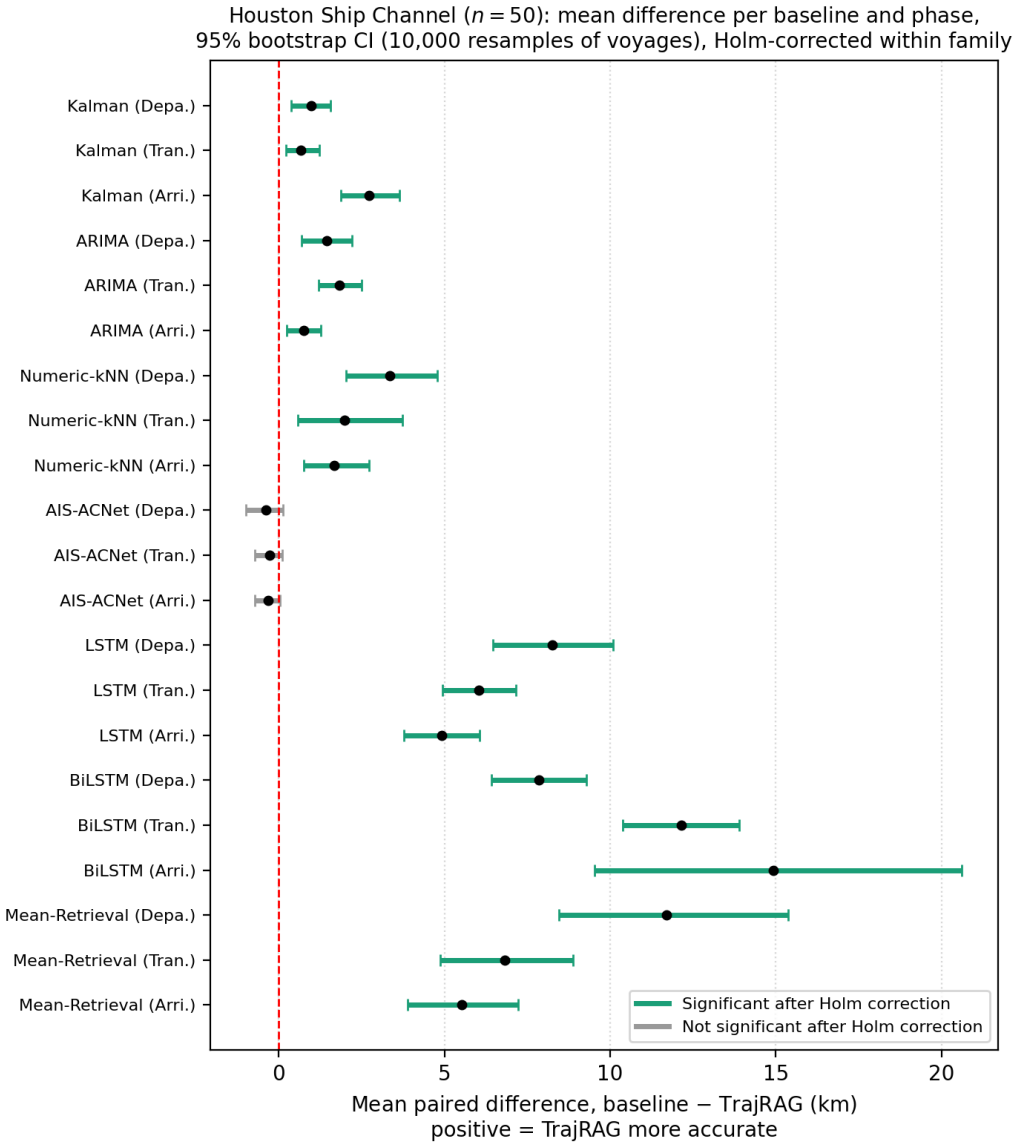


Figure 3: Mean paired difference (baseline − TrajRAG, km; positive = TrajRAG more accurate) with 95% bootstrap confidence interval (10,000 resamples of voyage identities), Houston Ship Channel, $n = 50$, deployable speed rule. Green: significant after Holm correction within this 21-test family; grey: not significant. This figure reports the mean difference and its uncertainty; Table 15 reports the corresponding median-difference interval and Table 8 the exact mean errors themselves.

Table 12: Paired two-sided Wilcoxon signed-rank tests, TrajRAG vs. each method, Los Angeles/Long Beach ($n = 10$). p_{Holm}: Holm-corrected within the 18-test family of this corridor. Bold: not significant at the 0.05 level after correction. Numeric-kNN was never rescored for this corridor (Section 7.4) and is omitted, not invented.

TrajRAG vs.	Departure		Transit		Arrival	
	p	p_{Holm}	p	p_{Holm}	p	p_{Holm}
Kalman	0.049	**0.342**	0.049	**0.342**	0.014	**0.123**
ARIMA	0.049	**0.342**	0.010	**0.098**	0.232	**0.697**
LSTM	2.0×10^{-3}	0.035	2.0×10^{-3}	0.035	2.0×10^{-3}	0.035
BiLSTM	2.0×10^{-3}	0.035	0.131	**0.523**	2.0×10^{-3}	0.035
Mean-Retrieval	2.0×10^{-3}	0.035	0.014	**0.123**	2.0×10^{-3}	0.035
AIS-ACNet	0.275	**0.697**	0.432	**0.697**	2.0×10^{-3}	0.035

Table 13: Paired two-sided Wilcoxon signed-rank test, TrajRAG vs. AIS-ACNet, short-horizon matched subset (Table 7). p_{Holm}: Holm-corrected within this 3-test family.

	Departure ($n = 63$)	Transit ($n = 63$)	Arrival ($n = 53$)
p	3.4×10^{-3}	7.4×10^{-8}	1.5×10^{-4}
p_{Holm}	3.4×10^{-3}	2.2×10^{-7}	3.0×10^{-4}

Table 14: Paired two-sided Wilcoxon signed-rank tests of TrajRAG (deployable speed rule) against each full-horizon method, matched on voyage identity ($n = 50$, Houston). p_{Holm}: Holm-corrected within this 21-test family (Section 6.7). Bold: not significant at the 0.05 level after correction.

TrajRAG vs.	Departure		Transit		Arrival	
	p	p_{Holm}	p	p_{Holm}	p	p_{Holm}
Kalman	9.5×10^{-5}	8.5×10^{-4}	2.9×10^{-4}	2.0×10^{-3}	1.1×10^{-10}	$< 10^{-3}$
ARIMA	1.2×10^{-4}	9.5×10^{-4}	8.4×10^{-8}	1.0×10^{-6}	3.2×10^{-4}	2.0×10^{-3}
LSTM	8.8×10^{-12}	$< 10^{-3}$	3.4×10^{-14}	$< 10^{-3}$	1.5×10^{-11}	$< 10^{-3}$
BiLSTM	1.1×10^{-12}	$< 10^{-3}$	1.2×10^{-14}	$< 10^{-3}$	8.8×10^{-12}	$< 10^{-3}$
Mean-Retrieval	5.3×10^{-15}	$< 10^{-3}$	8.0×10^{-10}	$< 10^{-3}$	3.1×10^{-9}	$< 10^{-3}$
Numeric-kNN	7.5×10^{-6}	7.5×10^{-5}	8.2×10^{-4}	4.1×10^{-3}	6.3×10^{-3}	2.5×10^{-2}
AIS-ACNet	**0.752**	**1.000**	**0.540**	**1.000**	**0.496**	**1.000**

Table 15: Median paired difference (baseline − TrajRAG, deployable speed rule, km) with 95% bootstrap confidence interval (10,000 resamples of voyages), $n = 50$ Houston. Bold: interval includes zero.

Method	Departure	Transit	Arrival
Kalman	0.86 [0.29, 1.42]	0.43 [0.16, 0.57]	1.32 [0.66, 3.62]
ARIMA	0.99 [0.53, 2.07]	1.43 [1.02, 2.29]	0.38 [0.11, 0.81]
LSTM	9.12 [5.31, 10.57]	5.11 [4.12, 7.40]	4.28 [2.85, 6.03]
BiLSTM	7.92 [6.96, 9.80]	12.85 [9.81, 15.02]	5.33 [2.40, 8.26]
Mean-Retrieval	6.12 [4.11, 11.44]	5.21 [2.72, 7.81]	4.13 [3.34, 6.97]
Numeric-kNN	2.35 [1.66, 3.81]	0.91 [0.54, 1.64]	0.30 [0.07, 0.87]
AIS-ACNet	**0.11 [−0.35, 0.30]**	**0.07 [−0.31, 0.23]**	**0.03 [−0.37, 0.38]**

Table 16: Mean Haversine error (km) by corridor: 50 Houston voyages, 10 Los Angeles/Long Beach voyages, deployable speed rule for TrajRAG in both corridors. AIS-ACNet's LA/Long Beach model uses a corridor-specific bounding box and window/block configuration (9/3, 7/3, 22/7 steps for departure/transit/arrival), distinct from Houston's (21/7, 15/5, 32/11).

Method	Houston Ship Channel ($n = 50$)			LA / Long Beach ($n = 10$)		
	Dep.	Trans.	Arr.	Dep.	Trans.	Arr.
Kalman	3.16	2.61	4.23	1.20	2.63	0.36
ARIMA	3.64	3.78	2.24	1.52	3.01	0.49
LSTM	10.44	7.99	6.42	4.69	5.01	4.24
BiLSTM	10.03	14.10	16.43	3.43	2.81	8.29
Mean-Retrieval	13.89	8.76	7.03	5.68	3.89	4.16
AIS-ACNet	1.80	1.66	1.18	1.13	1.95	1.02
TrajRAG	2.19	1.95	1.50	0.81	1.71	0.11

Table 17: Empirical displacement ratio $\rho = d_i/(v_{i-1}\Delta t)$ on the corrected Houston Ship Channel training split (test voyages and steps with near-zero SOG excluded), compared with the aviation calibration target. $Q_{0.99}$: 99th percentile.

Phase	Step pairs	$Q_{0.99}(\rho)$	Share with $\rho > 1.2$
Departure	33,821	1.95	11.98%
Transit	22,087	1.47	6.67%
Arrival	33,169	1.51	6.72%
All maritime phases	89,077	1.72	8.71%
Aviation, all phases (Felendler et al., 2026)	–	1.18	–

Table 18: Maritime ablations, 20 Houston test voyages, mean Haversine error (km), deployable speed rule. p: paired two-sided Wilcoxon signed-rank test against the full pipeline, matched on the same 20 voyages. p_{Holm}: Holm-corrected within this 9-test family. Bold: not significant at the 0.05 level after correction.

Configuration	Departure			Transit			Arrival		
	Mean	p	p_{Holm}	Mean	p	p_{Holm}	Mean	p	p_{Holm}
Full pipeline	2.56	–	–	1.63	–	–	1.47	–	–
w/o retrieval	2.59	**0.927**	**1.00**	2.42	0.012	**0.109**	2.20	0.058	**0.466**
w/o guard	3.23	**0.202**	**1.00**	1.59	**0.312**	**1.00**	1.97	**0.723**	**1.00**
w/o heading	2.60	**0.312**	**1.00**	1.47	**0.904**	**1.00**	1.78	**0.546**	**1.00**

Table 19: Maritime TrajRAG under the aviation engine's backbone and retrieval size (`gpt-5.1`, $K = 5$) vs. the maritime default (`gpt-4.1`, $K = 6$), 125 Houston test voyages, deployable speed rule. Mean Haversine error (km).

Backbone	Departure	Transit	Arrival
`gpt-4.1`, $K = 6$ (maritime default)	2.21	1.18	1.52
`gpt-5.1`, $K = 5$ (aviation settings)	**1.87**	1.30	1.73

Table 20: Operational budget of the shared engine in both domains. Aviation latency is per reconstructed state, excluding the one-off textual summary built at onset (Felendler et al., 2026); maritime latency and cost are per complete phase-level blackout reconstruction (mean over the 50-voyage Houston test set, deployable speed rule). Cost is the actual measured LLM API cost in US dollars (gpt-4.1 at $2.10 per million tokens).

Domain	Phase	Latency (s)	Cost (USD)	Reporting interval (s)
Aviation, per state	Takeoff / cruise / landing	2.0–2.8	0.014–0.018	120
Maritime, per blackout	Departure	18.1	0.149	300
	Transit	20.8	0.171	300
	Arrival	34.6	0.453	300

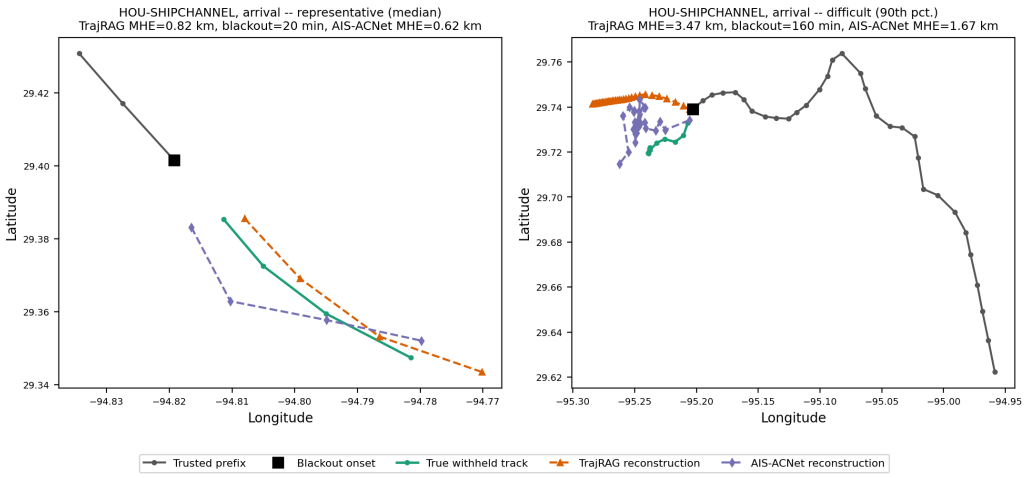


Figure 4: Houston Ship Channel, arrival phase. Left: representative case (voyage closest to the group median TrajRAG error, 0.82 km, 20-minute blackout). Right: difficult case (voyage closest to the group 90th percentile, 3.47 km, 160-minute blackout), where TrajRAG continues past the true track's turn into the berth approach while AIS-ACNet oscillates but stays closer to the true position – an instance of the heavy right tail noted in Section 6.5.

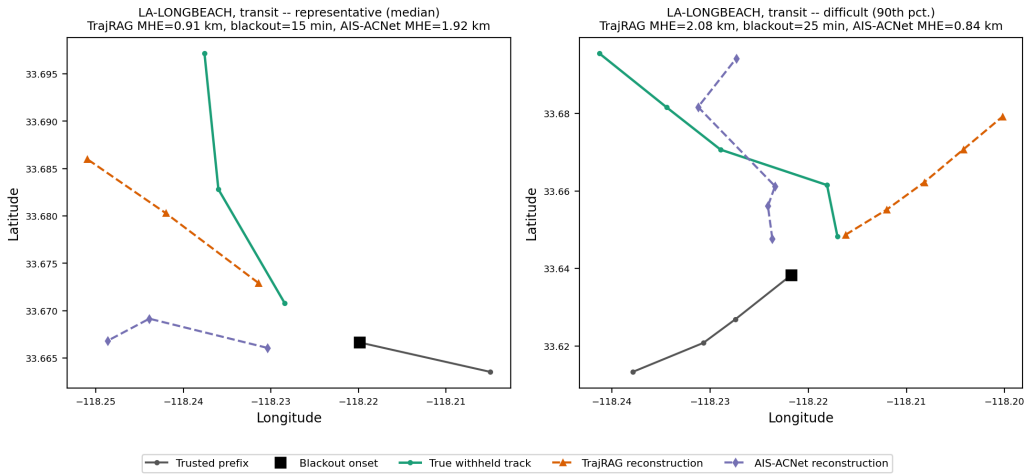


Figure 5: Los Angeles/Long Beach, transit phase. Left: representative case (0.91 km TrajRAG error, 15-minute blackout), where TrajRAG tracks the true withheld course more closely than AIS-ACNet. Right: difficult case (2.08 km TrajRAG error, 25-minute blackout), where AIS-ACNet's shorter, more conservative reconstruction stays closer to the true track than TrajRAG's. Together the two panels illustrate that neither method dominates the other case by case, consistent with the non-significant paired comparison for this corridor and phase (Table 12).